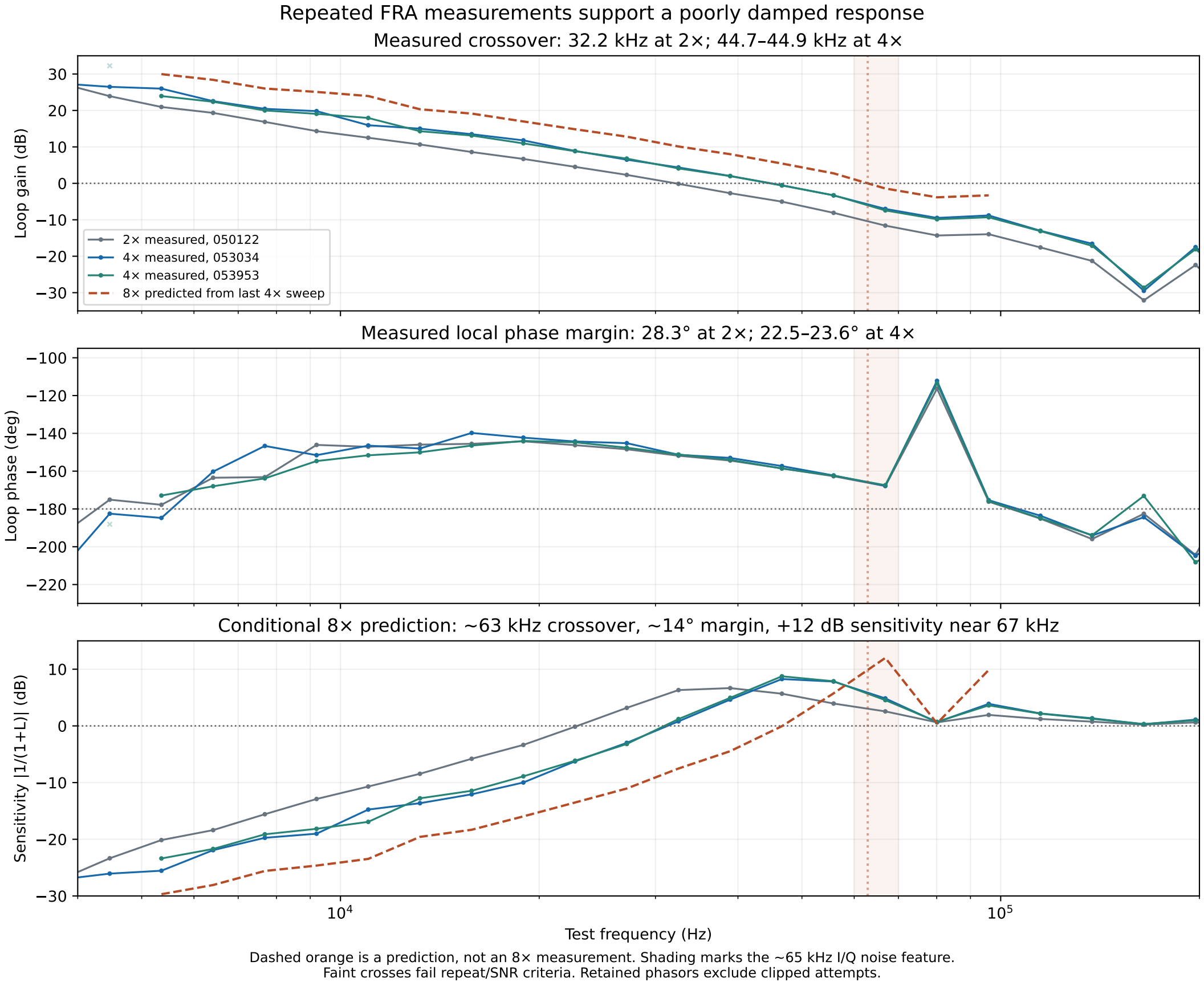
Repeated FRA measurements support a poorly damped response
Measured crossover: 32.2 kHz at 2×; 44.7–44.9 kHz at 4×
Loop gain (dB)
2× measured, 050122
4× measured, 053034
4× measured, 053953
8× predicted from last 4× sweep
Measured local phase margin: 28.3° at 2×; 22.5–23.6° at 4×
Loop phase (deg)
Conditional 8× prediction: ~63 kHz crossover, ~14° margin, +12 dB sensitivity near 67 kHz
Sensitivity |1/(1+L)| (dB)
Test frequency (Hz)
Dashed orange is a prediction, not an 8× measurement. Shading marks the ~65 kHz I/Q noise feature.
Faint crosses fail repeat/SNR criteria. Retained phasors exclude clipped attempts.